

**Microsoft
CERTIFIED**

**Fabric Analytics
Engineer Associate**





**Microsoft Fabric Analytics Engineer-
(DP 600)**

IMPLEMENTING ANALYTICS SOLUTIONS USING MICROSOFT FABRIC

As a candidate for this exam, you should have subject matter expertise in designing, creating, and deploying enterprise-scale data analytics solutions.

Your responsibilities for this role include transforming data into reusable analytics assets by using Microsoft Fabric components, such as:

Lakehouses

Data warehouses

Notebooks

Dataflows

Data pipelines

Semantic models

Reports

You implement analytics best practices in Fabric, including version control and deployment.

To implement solutions as a Fabric analytics engineer, you partner with other roles, such as:

Solution architects

Data engineers

Data scientists

AI engineers

Database administrators

Power BI data analysts

In addition to in-depth work with the Fabric platform, you need experience with:

Data modeling

Data transformation

Git-based source control

Exploratory analytics

Languages, including Structured Query Language (SQL), Data Analysis Expressions (DAX), and PySpark

Skills at a glance

Plan, implement, and manage a solution for data analytics (10–15%)

Prepare and serve data (40–45%)

Implement and manage semantic models (20–25%)

Explore and analyze data (20–25%)

Plan, implement, and manage a solution for data analytics (10–15%)

Plan a data analytics environment

Identify requirements for a solution, including components, features, performance, and capacity stock-keeping units (SKUs)

Recommend settings in the Fabric admin portal

Choose a data gateway type

Create a custom Power BI report theme

Implement and manage a data analytics environment

Implement workspace and item-level access controls for Fabric items

Implement data sharing for workspaces, warehouses, and lakehouses

Manage sensitivity labels in semantic models and lakehouses

Configure Fabric-enabled workspace settings

Manage Fabric capacity

Manage the analytics development lifecycle

Implement version control for a workspace

Create and manage a Power BI Desktop project (.pbip)

Plan and implement deployment solutions

Perform impact analysis of downstream dependencies from lakehouses, data warehouses, dataflows, and semantic models

Deploy and manage semantic models by using the XMLA endpoint

Create and update reusable assets, including Power BI template (.pbit) files, Power BI data source (.pbids) files, and shared semantic models

Prepare and serve data (40–45%)

Create objects in a lakehouse or warehouse

Ingest data by using a data pipeline, dataflow, or notebook

Create and manage shortcuts

Implement file partitioning for analytics workloads in a lakehouse

Create views, functions, and stored procedures

Enrich data by adding new columns or tables

Copy data

Choose an appropriate method for copying data from a Fabric data source to a lakehouse or warehouse

Copy data by using a data pipeline, dataflow, or notebook

Add stored procedures, notebooks, and dataflows to a data pipeline

Schedule data pipelines

Schedule dataflows and notebooks

Transform data

Implement a data cleansing process

Implement a star schema for a lakehouse or warehouse, including Type 1

and Type 2 slowly changing dimensions

Implement bridge tables for a lakehouse or a warehouse

Denormalize data

Aggregate or de-aggregate data

Merge or join data

Identify and resolve duplicate data, missing data, or null values

Convert data types by using SQL or PySpark

Filter data

Optimize performance

Identify and resolve data loading performance bottlenecks in dataflows, notebooks, and SQL queries

Implement performance improvements in dataflows, notebooks, and SQL queries

Identify and resolve issues with Delta table file sizes

Implement and manage semantic models (20–25%)

Design and build semantic models

Choose a storage mode, including Direct Lake

Identify use cases for DAX Studio and Tabular Editor 2

Implement a star schema for a semantic model

Implement relationships, such as bridge tables and many-to-many relationships

Write calculations that use DAX variables and functions, such as iterators, table filtering, windowing, and information functions

Implement calculation groups, dynamic strings, and field parameters

Design and build a large format dataset

Design and build composite models that include aggregations

Implement dynamic row-level security and object-level security

Validate row-level security and object-level security

Optimize enterprise-scale semantic models

Implement performance improvements in queries and report visuals

Improve DAX performance by using DAX Studio

Optimize a semantic model by using Tabular Editor 2

Implement incremental refresh

Explore and analyze data (20–25%)

Perform exploratory analytics

Implement descriptive and diagnostic analytics

Integrate prescriptive and predictive analytics into a visual or report

Profile data

Query data by using SQL

Query a lakehouse in Fabric by using SQL queries or the visual query editor

Query a warehouse in Fabric by using SQL queries or the visual query editor

Connect to and query datasets by using the XMLA endpoint

Case study - 1

Overview -

Contoso, Ltd. is a US-based health supplements company. Contoso has two divisions named Sales and Research. The Sales division contains two departments named Online Sales and Retail Sales. The Research division assigns internally developed product lines to individual teams of researchers and analysts.

Existing Environment -

Identity Environment -

Contoso has a Microsoft Entra tenant named contoso.com. The tenant contains two groups named ResearchReviewersGroup1 and ResearchReviewersGroup2.

Data Environment -

Contoso has the following data environment:

The Sales division uses a Microsoft Power BI Premium capacity.

The semantic model of the Online Sales department includes a fact table named Orders that uses Import mode. In the system of origin, the OrderID value represents the sequence in which orders are created.

The Research department uses an on-premises, third-party data warehousing product.

Fabric is enabled for contoso.com.

An Azure Data Lake Storage Gen2 storage account named storage1 contains Research division data for a product line named Productline1. The data is in the delta format.

A Data Lake Storage Gen2 storage account named storage2 contains Research division data for a product line named Productline2. The data is in the CSV format.

Requirements -

Planned Changes -

Contoso plans to make the following changes:

Enable support for Fabric in the Power BI Premium capacity used by the Sales division.

Make all the data for the Sales division and the Research division available in Fabric.

For the Research division, create two Fabric workspaces named Productline1ws and Productine2ws.

In Productline1ws, create a lakehouse named Lakehouse1.

In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

All the workspaces for the Sales division and the Research division must support all Fabric experiences.

The Research division workspaces must use a dedicated, on-demand capacity that has per-minute billing.

The Research division workspaces must be grouped together logically to support OneLake data hub filtering based on the department name.

For the Research division workspaces, the members of ResearchReviewersGroup1 must be able to read lakehouse and warehouse data and shortcuts by using SQL endpoints.

For the Research division workspaces, the members of ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.

All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -

Contoso identifies the following data preparation requirements:

The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.

All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -

Contoso identifies the following requirements for implementing and managing semantic models:

The number of rows added to the Orders table during refreshes must be minimized.

The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be

considered for all solutions:

Follow the principle of least privilege when applicable.

Minimize implementation and maintenance effort when possible.

You need to ensure that Contoso can use version control to meet the data analytics requirements and the general requirements.

What should you do?

A. Store at the semantic models and reports in Data Lake Gen2 storage.

B. Modify the settings of the Research workspaces to use a GitHub repository.

C. Modify the settings of the Research division workspaces to use an Azure Repos repository.

D. Store all the semantic models and reports in Microsoft OneDrive.

Correct Answer: B

Case study - 2

Overview -

Contoso, Ltd. is a US-based health supplements company. Contoso has two divisions named Sales and Research. The Sales division contains two departments named Online Sales and Retail Sales. The Research division assigns internally developed product lines to individual teams of researchers and analysts.

Existing Environment -

Identity Environment -

Contoso has a Microsoft Entra tenant named contoso.com. The tenant contains two groups named ResearchReviewersGroup1 and ResearchReviewersGroup2.

Data Environment -

Contoso has the following data environment:

The Sales division uses a Microsoft Power BI Premium capacity.

The semantic model of the Online Sales department includes a fact table named Orders that uses Import mode. In the system of origin, the OrderID value represents the sequence in which orders are created.

The Research department uses an on-premises, third-party data warehousing product.

Fabric is enabled for contoso.com.

An Azure Data Lake Storage Gen2 storage account named storage1 contains Research division data for a product line named Productline1. The data is in the delta format.

A Data Lake Storage Gen2 storage account named storage2 contains Research division data for a product line named Productline2. The data is in the CSV format.

Requirements -

Planned Changes -

Contoso plans to make the following changes:

Enable support for Fabric in the Power BI Premium capacity used by the Sales division.

Make all the data for the Sales division and the Research division available in Fabric.

For the Research division, create two Fabric workspaces named Productline1ws and Productline2ws.

In Productline1ws, create a lakehouse named Lakehouse1.

In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

All the workspaces for the Sales division and the Research division must support all Fabric experiences.

The Research division workspaces must use a dedicated, on-demand capacity that has per-minute billing.

The Research division workspaces must be grouped together logically to support OneLake data hub filtering based on the department name.

For the Research division workspaces, the members of ResearchReviewersGroup1 must be able to read lakehouse and warehouse data and shortcuts by using SQL endpoints.

For the Research division workspaces, the members of ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.

All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -

Contoso identifies the following data preparation requirements:

The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.

All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -

Contoso identifies the following requirements for implementing and

managing semantic models:

The number of rows added to the Orders table during refreshes must be minimized.

The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be considered for all solutions:

Follow the principle of least privilege when applicable.

Minimize implementation and maintenance effort when possible.

You need to recommend a solution to group the Research division workspaces.

What should you include in the recommendation?

Answer Area

Grouping method:

	▼
Capacity	
Domain	
Tenant	

Tool:

	▼
OneLake data hub	
The Fabric Admin portal	
The Microsoft Entra admin center	

Correct Answer:

Answer Area

Grouping method:

	▼
Capacity	
Domain	
Tenant	

Tool:

	▼
OneLake data hub	
The Fabric Admin portal	
The Microsoft Entra admin center	

Case study - 3

Overview -

Contoso, Ltd. is a US-based health supplements company. Contoso has two divisions named Sales and Research. The Sales division contains two departments named Online Sales and Retail Sales. The Research division assigns internally developed product lines to individual teams of researchers and analysts.

Existing Environment -

Identity Environment -

Contoso has a Microsoft Entra tenant named contoso.com. The tenant contains two groups named ResearchReviewersGroup1 and ResearchReviewersGroup2.

Data Environment -

Contoso has the following data environment:

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Fabric is enabled for contoso.com.

An Azure Data Lake Storage Gen2 storage account named storage1 contains Research division data for a product line named Productline1. The data is in the delta format.

A Data Lake Storage Gen2 storage account named storage2 contains

Research division data for a product line named Productline2. The data is in the CSV format.

Requirements -

Planned Changes -

Contoso plans to make the following changes:

Enable support for Fabric in the Power BI Premium capacity used by the Sales division.

Make all the data for the Sales division and the Research division available in Fabric.

For the Research division, create two Fabric workspaces named Productline1ws and Productine2ws.

In Productline1ws, create a lakehouse named Lakehouse1.

In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

All the workspaces for the Sales division and the Research division must support all Fabric experiences.

The Research division workspaces must use a dedicated, on-demand capacity that has per-minute billing.

The Research division workspaces must be grouped together logically to support OneLake data hub filtering based on the department name.

For the Research division workspaces, the members of ResearchReviewersGroup1 must be able to read lakehouse and warehouse data and shortcuts by using SQL endpoints.

For the Research division workspaces, the members of

ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.

All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -

Contoso identifies the following data preparation requirements:

The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.

All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -

Contoso identifies the following requirements for implementing and managing semantic models:

The number of rows added to the Orders table during refreshes must be minimized.

The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be considered for all solutions:

Follow the principle of least privilege when applicable.

Minimize implementation and maintenance effort when possible.

You need to refresh the Orders table of the Online Sales department. The solution must meet the semantic model requirements.

What should you include in the solution?

A. an Azure Data Factory pipeline that executes a Stored procedure activity to retrieve the maximum value of the OrderID column in the destination lakehouse

B. an Azure Data Factory pipeline that executes a Stored procedure activity to retrieve the minimum value of the OrderID column in the destination lakehouse

C. an Azure Data Factory pipeline that executes a dataflow to retrieve the minimum value of the OrderID column in the destination lakehouse

D. an Azure Data Factory pipeline that executes a dataflow to retrieve the maximum value of the OrderID column in the destination lakehouse

Correct Answer: D

Case study - 4

Overview -

Contoso, Ltd. is a US-based health supplements company. Contoso has two divisions named Sales and Research. The Sales division contains two departments named Online Sales and Retail Sales. The Research division assigns internally developed product lines to individual teams of researchers and analysts.

Existing Environment -

Identity Environment -

Contoso has a Microsoft Entra tenant named contoso.com. The tenant contains two groups named ResearchReviewersGroup1 and ResearchReviewersGroup2.

Data Environment -

Contoso has the following data environment:

The Sales division uses a Microsoft Power BI Premium capacity.

The semantic model of the Online Sales department includes a fact table named Orders that uses Import mode. In the system of origin, the OrderID value represents the sequence in which orders are created.

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Fabric is enabled for contoso.com.

An Azure Data Lake Storage Gen2 storage account named storage1 contains Research division data for a product line named Productline1. The data is in the delta format.

A Data Lake Storage Gen2 storage account named storage2 contains Research division data for a product line named Productline2. The data is in the CSV format.

Requirements -

Planned Changes -

Contoso plans to make the following changes:

Enable support for Fabric in the Power BI Premium capacity used by the Sales division.

Make all the data for the Sales division and the Research division available in Fabric.

For the Research division, create two Fabric workspaces named Productline1ws and Productine2ws.

In Productline1ws, create a lakehouse named Lakehouse1.

In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

All the workspaces for the Sales division and the Research division must support all Fabric experiences.

The Research division workspaces must use a dedicated, on-demand capacity that has per-minute billing.

The Research division workspaces must be grouped together logically to

support OneLake data hub filtering based on the department name.

For the Research division workspaces, the members of ResearchReviewersGroup1 must be able to read lakehouse and warehouse data and shortcuts by using SQL endpoints.

For the Research division workspaces, the members of ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.

All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -

Contoso identifies the following data preparation requirements:

The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.

All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -

Contoso identifies the following requirements for implementing and managing semantic models:

The number of rows added to the Orders table during refreshes must be minimized.

The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be considered for all solutions:

Follow the principle of least privilege when applicable.

Minimize implementation and maintenance effort when possible.

Which syntax should you use in a notebook to access the Research division data for Productline1?

A. `spark.read.format("delta").load("Tables/productline1/ResearchProduct")`

B. `spark.sql("SELECT * FROM Lakehouse1.ResearchProduct ")`

C. `external_table('Tables/ResearchProduct)`

D. `external_table(ResearchProduct)`

Correct Answer: A

Case study - 5

Overview -

Litware, Inc. is a manufacturing company that has offices throughout North America. The analytics team at Litware contains data engineers, analytics engineers, data analysts, and data scientists.

Existing Environment -

Fabric Environment -

Litware has been using a Microsoft Power BI tenant for three years. Litware has NOT enabled any Fabric capacities and features.

Available Data -

Litware has data that must be analyzed as shown in the following table.

Description	Original source	Total size
Customer data	Customer relationship management (CRM) system	50 MB
Product data	Customer relationship management (CRM) system	200 MB
Customer satisfaction surveys	SurveyMonkey	500 GB

The Product data contains a single table and the following columns.

Name	Data type
ProductID	Integer
ProductName	String
ProductCategory	String
ListPrice	Decimal

The customer satisfaction data contains the following tables:

Survey -

Question -

Response -

For each survey submitted, the following occurs:

One row is added to the Survey table.

One row is added to the Response table for each question in the survey.

The Question table contains the text of each survey question. The third question in each survey response is an overall satisfaction score. Customers can submit a survey after each purchase.

User Problems -

The analytics team has large volumes of data, some of which is semi-structured. The team wants to use Fabric to create a new data store.

Product data is often classified into three pricing groups: high, medium, and low. This logic is implemented in several databases and semantic models, but the logic does NOT always match across implementations.

Requirements -

Planned Changes -

Litware plans to enable Fabric features in the existing tenant. The analytics team will create a new data store as a proof of concept (PoC). The remaining Liware users will only get access to the Fabric features once the PoC is complete. The PoC will be completed by using a Fabric trial capacity

The following three workspaces will be created:

AnalyticsPOC: Will contain the data store, semantic models, reports pipelines, dataflow, and notebooks used to populate the data store

DataEngPOC: Will contain all the pipelines, dataflows, and notebooks used to populate OneLake

DataSciPOC: Will contain all the notebooks and reports created by the data scientists

The following will be created in the AnalyticsPOC workspace:

A data store (type to be decided)

A custom semantic model -

A default semantic model -

Interactive reports -

The data engineers will create data pipelines to load data to OneLake either hourly or daily depending on the data source. The analytics engineers will create processes to ingest, transform, and load the data to the data store in the AnalyticsPOC workspace daily. Whenever possible, the data engineers will use low-code tools for data ingestion. The choice of which data

cleansing and transformation tools to use will be at the data engineers' discretion.

All the semantic models and reports in the Analytics POC workspace will use the data store as the sole data source.

Technical Requirements -

The data store must support the following:

Read access by using T-SQL or Python

Semi-structured and unstructured data

Row-level security (RLS) for users executing T-SQL queries

Files loaded by the data engineers to OneLake will be stored in the Parquet format and will meet Delta Lake specifications.

Data will be loaded without transformation in one area of the AnalyticsPOC data store. The data will then be cleansed, merged, and transformed into a dimensional model

The data load process must ensure that the raw and cleansed data is updated completely before populating the dimensional model

The dimensional model must contain a date dimension. There is no existing data source for the date dimension. The Litware fiscal year matches the calendar year. The date dimension must always contain dates from 2010 through the end of the current year.

The product pricing group logic must be maintained by the analytics engineers in a single location. The pricing group data must be made available in the data store for T-SQL queries and in the default semantic model. The following logic must be used:

List prices that are less than or equal to 50 are in the low pricing group.

List prices that are greater than 50 and less than or equal to 1,000 are in the medium pricing group.

List prices that are greater than 1,000 are in the high pricing group.

Security Requirements -

Only Fabric administrators and the analytics team must be able to see the Fabric items created as part of the PoC.

Litware identifies the following security requirements for the Fabric items in the AnalyticsPOC workspace:

Fabric administrators will be the workspace administrators.

The data engineers must be able to read from and write to the data store. No access must be granted to datasets or reports.

The analytics engineers must be able to read from, write to, and create schemas in the data store. They also must be able to create and share semantic models with the data analysts and view and modify all reports in the workspace.

The data scientists must be able to read from the data store, but not write to it. They will access the data by using a Spark notebook

The data analysts must have read access to only the dimensional model objects in the data store. They also must have access to create Power BI reports by using the semantic models created by the analytics engineers.

The date dimension must be available to all users of the data store.

The principle of least privilege must be followed.

Both the default and custom semantic models must include only tables or views from the dimensional model in the data store. Litware already has the following Microsoft Entra security groups:

FabricAdmins: Fabric administrators

AnalyticsTeam: All the members of the analytics team

DataAnalysts: The data analysts on the analytics team

DataScientists: The data scientists on the analytics team

DataEngineers: The data engineers on the analytics team

AnalyticsEngineers: The analytics engineers on the analytics team

Report Requirements -

The data analysts must create a customer satisfaction report that meets the following requirements:

Enables a user to select a product to filter customer survey responses to only those who have purchased that product.

Displays the average overall satisfaction score of all the surveys submitted during the last 12 months up to a selected date.

Shows data as soon as the data is updated in the data store.

Ensures that the report and the semantic model only contain data from the current and previous year.

Ensures that the report respects any table-level security specified in the source data store.

Minimizes the execution time of report queries.

You need to assign permissions for the data store in the AnalyticsPOC workspace. The solution must meet the security requirements.

Which additional permissions should you assign when you share the data store?

Answer Area

DataEngineers:

	▼
Build Reports on the default dataset Read All Apache Spark Read All SQL analytics endpoint data	

DataAnalysts:

	▼
Build Reports on the default dataset Read All Apache Spark Read All SQL analytics endpoint data	

DataScientists:

	▼
Build Reports on the default dataset Read All Apache Spark Read All SQL analytics endpoint data	

Correct Answer:

Answer Area

DataEngineers: ▼

- Build Reports on the default dataset
- Read All Apache Spark
- Read All SQL analytics endpoint data

DataAnalysts: ▼

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- Read All Apache Spark
- Read All SQL analytics endpoint data

DataScientists: ▼

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Case study - 6

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The dimensional model must contain a date dimension. There is no existing data source for the date dimension. The Litware fiscal year matches the calendar year. The date dimension must always contain dates from 2010 through the end of the current year.

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The data analysts must have read access to only the dimensional model objects in the data store. They also must have access to create Power BI reports by using the semantic models created by the analytics engineers.

The date dimension must be available to all users of the data store.

The principle of least privilege must be followed.

Both the default and custom semantic models must include only tables or views from the dimensional model in the data store. Litware already has the following Microsoft Entra security groups:

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- Enables a user to select a product to filter customer survey responses to only those who have purchased that product.

- Displays the average overall satisfaction score of all the surveys submitted during the last 12 months up to a selected date.

- Shows data as soon as the data is updated in the data store.

- Ensures that the report and the semantic model only contain data from the current and previous year.

- Ensures that the report respects any table-level security specified in the source data store.

- Minimizes the execution time of report queries.

You need to create a DAX measure to calculate the average overall satisfaction score. How should you complete the DAX code?

Answer Area

```
Rolling 12 Overall Satisfaction =  
VAR NumberOfMonths = 12  
VAR LastCurrentDate = MAX ( 'Date' [Date] )  
VAR Period = DATESINPERIOD ( 'Date' [Date], LastCurrentDate,  
- NumberOfMonths, MONTH )  
VAR Result =  
    CALCULATE (  
        AVERAGE('Survey'[Response Value]),  
        AVERAGEA('Question'[Question Text]),  
        AVERAGEX(VALUES('Survey'[Customer Key]),  
            LastCurrentDate,  
            NumberOfMonths,  
            Period,  
            'Survey Question' [Question Title] = "Overall Satisfaction"  
        )  
    )  
RETURN  
    Result
```

Correct Answer:

Answer Area

```
Rolling 12 Overall Satisfaction =  
VAR NumberOfMonths = 12  
VAR LastCurrentDate = MAX ( 'Date' [Date] )  
VAR Period = DATESINPERIOD ( 'Date' [Date], LastCurrentDate,  
- NumberOfMonths, MONTH )  
VAR Result =  
    CALCULATE (  
        AVERAGE('Survey'[Response Value]),  
        AVERAGEA('Question'[Question Text]),  
        AVERAGEX(VALUES('Survey'[Customer Key]),  
        LastCurrentDate,  
        NumberOfMonths,  
        Period,  
        'Survey Question' [Question Title] = "Overall Satisfaction"  
    )  
    RETURN  
        Result
```

Case study - 7

Overview -

Litware, Inc. is a manufacturing company that has offices throughout North America. The analytics team at Litware contains data engineers, analytics

engineers, data analysts, and data scientists.

Existing Environment -

Fabric Environment -

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Available Data -

Litware has data that must be analyzed as shown in the following table.

Description	Original source	Total size
Customer data	Customer relationship management (CRM) system	50 MB
Product data	Customer relationship management (CRM) system	200 MB
Customer satisfaction surveys	SurveyMonkey	500 GB

The Product data contains a single table and the following columns.

Name	Data type
ProductID	Integer
ProductName	String
ProductCategory	String
ListPrice	Decimal

The customer satisfaction data contains the following tables:

Survey -

Question -

Response -

For each survey submitted, the following occurs:

One row is added to the Survey table.

One row is added to the Response table for each question in the survey.

The Question table contains the text of each survey question. The third question in each survey response is an overall satisfaction score. Customers can submit a survey after each purchase.

User Problems -

The analytics team has large volumes of data, some of which is semi-structured. The team wants to use Fabric to create a new data store.

Product data is often classified into three pricing groups: high, medium, and low. This logic is implemented in several databases and semantic models, but the logic does NOT always match across implementations.

Requirements -

Planned Changes -

Litware plans to enable Fabric features in the existing tenant. The analytics team will create a new data store as a proof of concept (PoC). The remaining Liware users will only get access to the Fabric features once the PoC is complete. The PoC will be completed by using a Fabric trial capacity

The following three workspaces will be created:

AnalyticsPOC: Will contain the data store, semantic models, reports pipelines, dataflow, and notebooks used to populate the data store

DataEngPOC: Will contain all the pipelines, dataflows, and notebooks used to populate OneLake

DataSciPOC: Will contain all the notebooks and reports created by the data scientists

The following will be created in the AnalyticsPOC workspace:

A data store (type to be decided)

A custom semantic model -

A default semantic model -

Interactive reports -

The data engineers will create data pipelines to load data to OneLake either hourly or daily depending on the data source. The analytics engineers will create processes to ingest, transform, and load the data to the data store in the AnalyticsPOC workspace daily. Whenever possible, the data engineers will use low-code tools for data ingestion. The choice of which data

cleansing and transformation tools to use will be at the data engineers' discretion.

All the semantic models and reports in the Analytics POC workspace will use the data store as the sole data source.

Technical Requirements -

The data store must support the following:

Read access by using T-SQL or Python

Semi-structured and unstructured data

Row-level security (RLS) for users executing T-SQL queries

Files loaded by the data engineers to OneLake will be stored in the Parquet format and will meet Delta Lake specifications.

Data will be loaded without transformation in one area of the AnalyticsPOC data store. The data will then be cleansed, merged, and transformed into a dimensional model

The data load process must ensure that the raw and cleansed data is updated completely before populating the dimensional model

The dimensional model must contain a date dimension. There is no existing data source for the date dimension. The Litware fiscal year matches the calendar year. The date dimension must always contain dates from 2010 through the end of the current year.

The product pricing group logic must be maintained by the analytics engineers in a single location. The pricing group data must be made available in the data store for T-SOL. queries and in the default semantic model. The following logic must be used:

List prices that are less than or equal to 50 are in the low pricing group.

List prices that are greater than 50 and less than or equal to 1,000 are in the medium pricing group.

List prices that are greater than 1,000 are in the high pricing group.

Security Requirements -

Only Fabric administrators and the analytics team must be able to see the Fabric items created as part of the PoC.

Litware identifies the following security requirements for the Fabric items in the AnalyticsPOC workspace:

Fabric administrators will be the workspace administrators.

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The analytics engineers must be able to read from, write to, and create schemas in the data store. They also must be able to create and share semantic models with the data analysts and view and modify all reports in the workspace.

The data scientists must be able to read from the data store, but not write to it. They will access the data by using a Spark notebook

The data analysts must have read access to only the dimensional model objects in the data store. They also must have access to create Power BI reports by using the semantic models created by the analytics engineers.

The date dimension must be available to all users of the data store.

The principle of least privilege must be followed.

Both the default and custom semantic models must include only tables or views from the dimensional model in the data store. Litware already has the following Microsoft Entra security groups:

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Report Requirements -

The data analysts must create a customer satisfaction report that meets the following requirements:

Enables a user to select a product to filter customer survey responses to only those who have purchased that product.

Displays the average overall satisfaction score of all the surveys submitted during the last 12 months up to a selected date.

Shows data as soon as the data is updated in the data store.

Ensures that the report and the semantic model only contain data from the current and previous year.

Ensures that the report respects any table-level security specified in the source data store.

Minimizes the execution time of report queries.

You need to resolve the issue with the pricing group classification. How should you complete the T-SQL statement?

Answer Area

CREATE [dbo].[ProductsWithPricingGroup]

FUNCTION
PROCEDURE
TABLE
VIEW

AS

SELECT ProductId,
ProductName,
ProductCategory,
ListPrice,

CASE
COALESCE
IIF
SET

WHEN ListPrice <= 50 THEN 'low'

WHEN(ListPrice > 50 AND ListPrice < 1000) THEN 'medium'
WHEN(ListPrice >= 50 AND ListPrice < 1000) THEN 'medium'
WHEN ListPrice BETWEEN 50 AND 1000) THEN 'medium'

WHEN ListPrice > 1000 THEN 'high'

END AS PricingGroup

FROM dbo.Products

Correct Answer:

Answer Area

CREATE  [dbo].[ProductsWithPricingGroup]

FUNCTION
PROCEDURE
TABLE
VIEW

AS

```
SELECT ProductId,  
       ProductName,  
       ProductCategory,  
       ListPrice,
```


CASE
COALESCE
IF
SET

```
    WHEN ListPrice <= 50 THEN 'low'
```


 WHEN(ListPrice > 50 AND ListPrice < 1000) THEN 'medium'
 WHEN(ListPrice >= 50 AND ListPrice < 1000) THEN 'medium'
 WHEN ListPrice BETWEEN 50 AND 1000) THEN 'medium'

```
    WHEN ListPrice > 1000 THEN 'high'
```

```
END AS PricingGroup  
FROM dbo.Products
```

Case study - 8

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What should you recommend using to ingest the customer data into the data store in the AnalyticsPOC workspace?

A. a stored procedure

B. a pipeline that contains a KQL activity

C. a Spark notebook

D. a dataflow

Correct Answer: D

Case study - 9

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Ensures that the report and the semantic model only contain data from the current and previous year.

Ensures that the report respects any table-level security specified in the source data store.

Minimizes the execution time of report queries.

Which type of data store should you recommend in the AnalyticsPOC workspace?

A. a data lake

B. a warehouse

C. a lakehouse

D. an external Hive metastore

Correct Answer: C

Question

You have a Fabric warehouse that contains a table named Staging.Sales. Staging.Sales contains the following columns.

Name	Data type	Nullable
ProductID	Integer	No
ProductName	Varchar(30)	No
SalesDate	Datetime2(6)	No
WholesalePrice	Decimal(18,2)	Yes
Amount	Decimal(18,2)	Yes

You need to write a T-SQL query that will return data for the year 2023 that displays ProductID and ProductName and has a summarized Amount that is higher than 10,000. Which query should you use?

A.

```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
WHERE DATEPART(YEAR,SaleDate) = '2023'
GROUP BY ProductID, ProductName
HAVING SUM(Amount) > 10000
```

B.

```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
GROUP BY ProductID, ProductName
HAVING DATEPART(YEAR,SaleDate) = '2023' AND SUM(Amount) > 10000
```

C.

```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
WHERE DATEPART(YEAR,SaleDate) = '2023' AND SUM(Amount) > 10000
```

D.

```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
WHERE DATEPART(YEAR,SaleDate) = '2023'
GROUP BY ProductID, ProductName
HAVING TotalAmount > 10000
```

Correct Answer: A

Question

You have a data warehouse that contains a table named Stage.Customers.

Stage.Customers contains all the customer record updates from a customer relationship management (CRM) system. There can be multiple updates per customer. You need to write a T-SQL query that will return the customer ID, name, postal code, and the last updated time of the most recent row for each customer ID. How should you complete the code?

Answer Area

```
WITH CUSTOMERBASE AS (  
    SELECT [CustomerID]  
    , [CustomerName]  
    , [PostalCode]  
    , [LastUpdated]  
    , x =  OVER (PARTITION BY CustomerID  
                                ORDER BY LastUpdated DESC)  
    FROM [LakehousePOC].[dbo].[CustomerChanges]  
    )  
    SELECT CustomerID, CustomerName, PostalCode, LastUpdated  
    FROM CUSTOMERBASE  
    
```

LAST_Value()
NTILE()
ROW_NUMBER()
Having Max(LastUpdated) = 1
WHERE LastUpdated = Max(LastUpdated)
WHERE X = 1

Correct Answer:

Box 1: ROW_NUMBER()

Box 2: WHERE X = 1

Question #

You have a Fabric tenant. You plan to create a Fabric notebook that will use Spark DataFrames to generate Microsoft Power BI visuals.

You run the following code.

```
from powerbiclient import QuickVisualize, get_dataset_config, Report

PBI_visualize = QuickVisualize(get_dataset_config(df))
PBI_visualize
```

For each of the following statements, select Yes if the statement is true.

Otherwise, select No.

Answer Area

Statements	Yes	No
The code embeds an existing Power BI report.	☐	☐
The code creates a Power BI report.	☐	☐
The code displays a summary of the DataFrame.	☐	☐

Correct Answer:

Answer Area

Statements	Yes	No
The code embeds an existing Power BI report.	☐	☒
The code creates a Power BI report.	☒	☐
The code displays a summary of the DataFrame.	☒	☐

Question

You are the administrator of a Fabric workspace that contains a lakehouse named Lakehouse1. Lakehouse1 contains the following tables:

Table1: A Delta table created by using a shortcut

Table2: An external table created by using Spark

Table3: A managed table -

You plan to connect to Lakehouse1 by using its SQL endpoint.
What will you be able to do after connecting to Lakehouse1?

- A. Read Table3.
- B. Update the data Table3.
- C. Read Table2.
- D. Update the data in Table1.

Correct Answer: D

Question #

You have a Fabric tenant that contains a warehouse. You use a dataflow to load a new dataset from OneLake to the warehouse.

You need to add a PowerQuery step to identify the maximum values for the numeric columns. Which function should you include in the step?

- A. Table.MaxN
- B. Table.Max
- C. Table.Range
- D. Table.Profile

Correct Answer: B

Question #

You have a Fabric tenant that contains a machine learning model registered in a Fabric workspace. You need to use the model to generate predictions by using the PREDICT function in a Fabric notebook. Which two languages can you use to perform model scoring?

- A. T-SQL
- B. DAX
- C. Spark SQL
- D. PySpark

Correct Answer: CD

Question #

You are analyzing the data in a Fabric notebook. You have a Spark DataFrame assigned to a variable named df.

You need to use the Chart view in the notebook to explore the data manually. Which function should you run to make the data available in the Chart view?

- A. displayHTML
- B. show
- C. write
- D. display

Correct Answer: D

Question #

You have a Fabric tenant that contains a Microsoft Power BI report named Report1. Report1 includes a Python visual. Data displayed by the visual is grouped automatically and duplicate rows are NOT displayed. You need all rows to appear in the visual. What should you do?

- A. Reference the columns in the Python code by index.
- B. Modify the Sort Column By property for all columns.
- C. Add a unique field to each row.
- D. Modify the Summarize By property for all columns.

Correct Answer: A

Question

You have a Fabric tenant that contains a semantic model. The model contains data about retail stores. You need to write a DAX query that will be executed by using the XMLA endpoint. The query must return a table of stores that have opened since December 1, 2023. How should you complete the DAX expression? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Values**Answer Area**

DEFINE

EVALUATE

FILTER

SUMMARIZE

TABLE

```
VAR_SalesSince =  
    DATE (2023, 12, 01)  
  
FILTER (  
    (Store, Store[Name], Store[OpenDate]),  
    Store[OpenDate] >= _SalesSince  
)
```

Correct Answer:

Answer Area

DEFINE

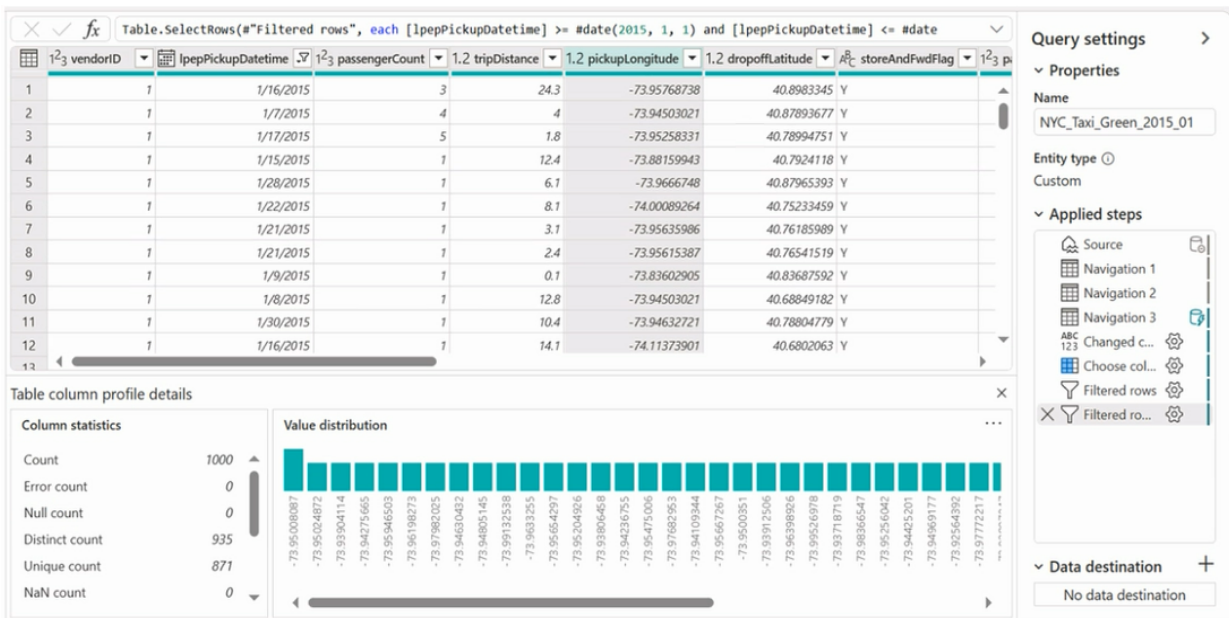
```
VAR_SalesSince =  
    DATE (2023, 12, 01)
```

EVALUATE

```
FILTER (  
    TABLE (Store, Store[Name], Store[OpenDate]),  
    Store[OpenDate] >= _SalesSince  
)
```

Question

You have a Fabric workspace named Workspace1 that contains a dataflow named Dataflow1. Dataflow1 has a query that returns 2,000 rows. You view the query in Power Query as shown in the following exhibit.



What can you identify about the pickupLongitude column?

- A. The column has duplicate values.
- B. All the table rows are profiled.
- C. The column has missing values.
- D. There are 935 values that occur only once.

Correct Answer: A

Question #

You have a Fabric tenant named Tenant1 that contains a workspace named WS1. WS1 uses a capacity named C1 and contains a dataset named DS1. You need to ensure read-write access to DS1 is available by using XMLA endpoint. What should be modified first?

- A. the DS1 settings
- B. the WS1 settings
- C. the C1 settings
- D. the Tenant1 settings

Correct Answer: C

Question #

You have a Fabric tenant that contains a workspace named Workspace1. Workspace1 is assigned to a Fabric capacity. You need to recommend a solution to provide users with the ability to create and publish custom Direct Lake semantic models by using external tools. The solution must follow the principle of least privilege. Which three actions in the Fabric Admin portal should you include in the recommendation?

- A. From the Tenant settings, set Allow XMLA Endpoints and Analyze in Excel with on-premises datasets to Enabled.
- B. From the Tenant settings, set Allow Azure Active Directory guest users to access Microsoft Fabric to Enabled.
- C. From the Tenant settings, select Users can edit data model in the Power BI service.
- D. From the Capacity settings, set XMLA Endpoint to Read Write.
- E. From the Tenant settings, set Users can create Fabric items to Enabled.
- F. From the Tenant settings, enable Publish to Web.

Correct Answer: ACD

Question

You are creating a semantic model in Microsoft Power BI Desktop. You plan to make bulk changes to the model by using the Tabular Model Definition

Language (TMDL) extension for Microsoft Visual Studio Code.

You need to save the semantic model to a file.

Which file format should you use?

- A. PBIP
- B. PBIX
- C. PBIT
- D. PBIDS

Correct Answer: B

Question #

You have a Fabric tenant that contains a warehouse named Warehouse1.

Warehouse1 contains three schemas named schemaA, schemaB, and schemaC. You need to ensure that a user named User1 can truncate tables in schemaA only. How should you complete the T-SQL statement?

Answer Area

GRANT	▼	ON	▼	TO User1;
	ALTER CONNECT EXECUTE		DATABASE::schemaA OBJECT::schemaA SCHEMA::schemaA	

Correct Answer:

Answer Area

GRANT ▼ ON ▼ TO User1;

ALTER	DATABASE::schemaA
CONNECT	OBJECT::schemaA
EXECUTE	SCHEMA::schemaA

Question

You plan to deploy Microsoft Power BI items by using Fabric deployment pipelines. You have a deployment pipeline that contains three stages named Development, Test, and Production.

A workspace is assigned to each stage. You need to provide Power BI developers with access to the pipeline. The solution must meet the following requirements:

Ensure that the developers can deploy items to the workspaces for Development and Test.

Prevent the developers from deploying items to the workspace for Production.

Follow the principle of least privilege.

Which three levels of access should you assign to the developers?

- A. Build permission to the production semantic models
- B. Admin access to the deployment pipeline
- C. Viewer access to the Development and Test workspaces
- D. Viewer access to the Production workspace
- E. Contributor access to the Development and Test workspaces
- F. Contributor access to the Production workspace

Correct Answer: BDE

QUESTION

You have a Fabric workspace that contains a DirectQuery semantic model. The model queries a data source that has 500 million rows. You have a Microsoft Power Bi report named Report1 that uses the model. Report1 contains visuals on multiple pages. You need to reduce the query execution time for the visuals on all the pages. What are two features that you can use?

- A.user-defined aggregations
- B.automatic aggregation
- C.query caching
- D.OneLake integration

Answer(s): A,B

QUESTION

You have a Fabric tenant that contains 30 CSV files in OneLake. The files are updated daily. You create a Microsoft Power BI semantic model named Model1 that uses the CSV files as a data source. You configure incremental refresh for Model1 and publish the model to a Premium capacity in the Fabric tenant. When you initiate a refresh of Model1, the refresh fails after running out of resources.

What is a possible cause of the failure?

- A.Query folding is occurring.

B.Only refresh complete days is selected.

C.XMLA Endpoint is set to Read Only.

D.Query folding is NOT occurring.

E.The delta type of the column used to partition the data has changed.

Answer(s): D

QUESTION

You have a Fabric tenant that uses a Microsoft Power BI Premium capacity.
You need to enable scale-out for a semantic model.What should you do first?

A.At the semantic model level, set Large dataset storage format to Off.

B.At the tenant level, set Create and use Metrics to Enabled.

C.At the semantic model level, set Large dataset storage format to On.

D.At the tenant level, set Data Activator to Enabled.

Answer(s): C

QUESTION

You have a Fabric tenant that contains a warehouse. The warehouse uses row-level security (RLS). You create a Direct Lake semantic model that uses the Delta tables and RLS of the warehouse. When users interact with a report built from the model, which mode will be used by the DAX queries?

A.DirectQuery

B.Dual

C.Direct Lake

D.Import

Answer(s): A

QUESTION

You have a Fabric tenant that contains a complex semantic model. The model is based on a star schema and contains many tables, including a fact table named Sales. You need to create a diagram of the model. The diagram must contain only the Sales table and related tables. What should you use from Microsoft Power BI Desktop?

- A.data categories
- B.Data view
- C.Model view
- D.DAX query view

Answer(s): C

QUESTION

You have a Fabric tenant that contains a semantic model. The model uses Direct Lake mode. You suspect that some DAX queries load unnecessary columns into memory.

You need to identify the frequently used columns that are loaded into memory. What are two ways to achieve the goal?

- A.Use the Analyze in Excel feature.

B. Use the Vertipaq Analyzer tool.

C. Query the

\$System.DISCOVER_STORAGE_TABLE_COLUMN_SEGMENTS

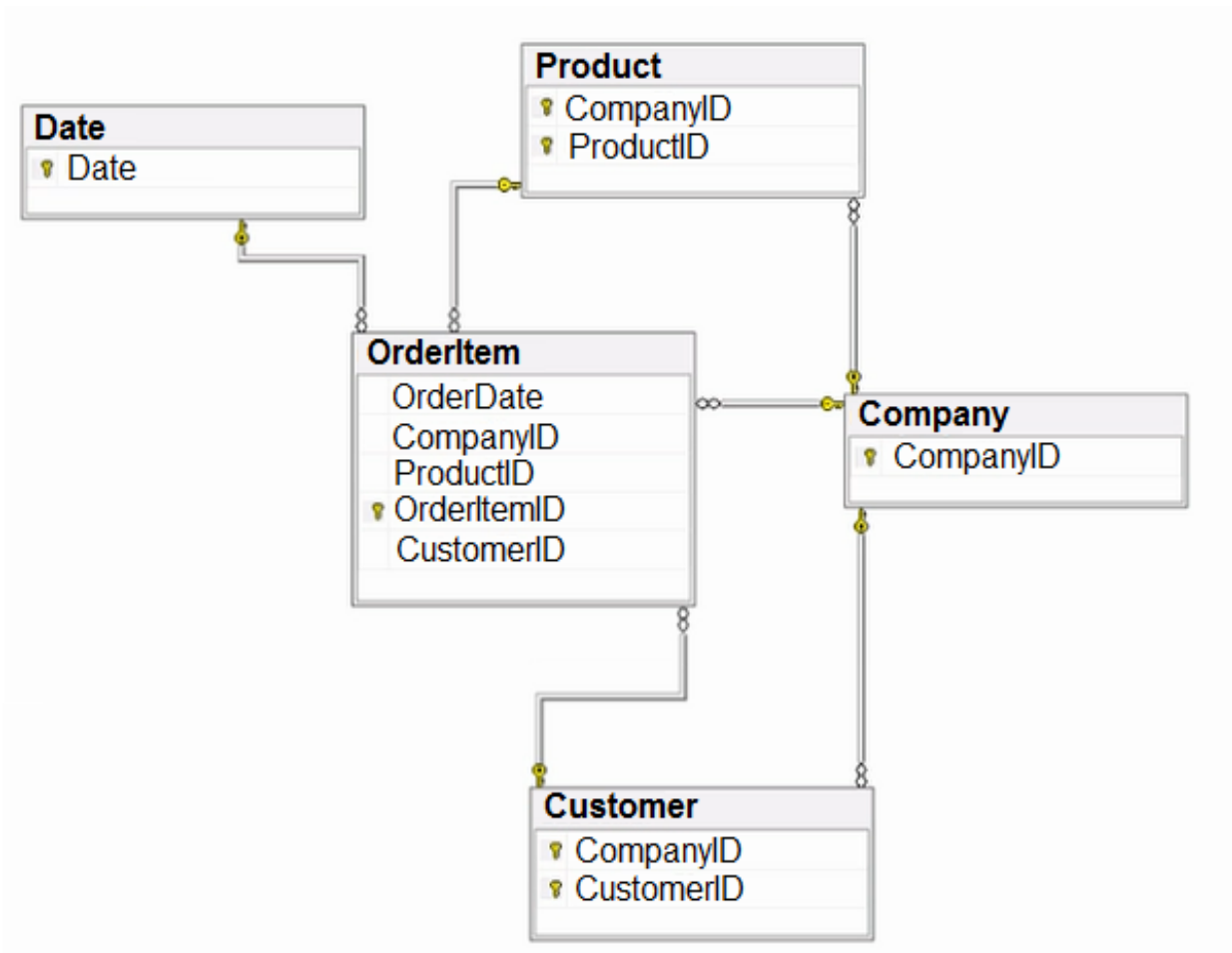
dynamic management view (DMV).

D. Query the DISCOVER_MEMORYGRANT dynamic management view (DMV).

Answer(s): B,C

QUESTION

You have the source data model shown in the following exhibit.



The primary keys of the tables are indicated by a key symbol beside the columns involved in each key.

You need to create a dimensional data model that will enable the analysis of order items by date, product, and customer.

What should you include in the solution?

Answer Area

The relationship between OrderItem and Product must be based on:

- The ProductID column
- Both the CompanyID and the ProductID columns
- A new key that combines the CompanyID and ProductID columns

The Company entity must be:

- Omitted
- Denormalized into the Product entity only
- Denormalized into the Customer and Product entities

Answer(s): A

Answer Area

The relationship between OrderItem and Product must be based on:

- The ProductID column
- Both the CompanyID and the ProductID columns
- A new key that combines the CompanyID and ProductID columns

The Company entity must be:

- Omitted
- Denormalized into the Product entity only
- Denormalized into the Customer and Product entities

QUESTION #

You have a Fabric tenant that contains a semantic model named Model1. Model1 uses Import mode. Model1 contains a table named Orders. Orders

has 100 million rows and the following fields.

Name	Data type	Description
OrderID	Integer	Column imported from the source
OrderDateTime	Date/Time	Column imported from the source
Quantity	Integer	Column imported from the source
Price	Decimal	Column imported from the source
TotalSalesAmount	Decimal	Calculated column that multiplies Quantity and Price
TotalQuantity	Integer	Measure

You need to reduce the memory used by Model1 and the time it takes to refresh the model. Which two actions should you perform?

A.Split OrderDateTime into separate date and time columns.

B.Replace TotalQuantity with a calculated column.

C.Convert Quantity into the Text data type.

D.Replace TotalSalesAmount with a measure.

Answer(s): A,D

QUESTION #

You have a Fabric tenant that contains a semantic model. You need to prevent report creators from populating visuals by using implicit measures. What are two tools that you can use to achieve the goal?

- A. Microsoft Power BI Desktop
- B. Tabular Editor
- C. Microsoft SQL Server Management Studio (SSMS)
- D. DAX Studio

Answer(s): A,B

QUESTION #

You have a Fabric tenant that contains two lakehouses. You are building a dataflow that will combine data from the lakehouses. The applied steps from one of the queries in the dataflow is shown in the following exhibit.

Query settings



▼ Properties

Name

Customers1

Entity type ⓘ

Custom

▼ Applied steps

	Source		
	Navigation 1		
	Capitalized each word		
	Appended query		
	Changed column type		
	Added custom		
	Filtered rows		

 Edit settings

 Rename

 Delete

 Delete until end

 Insert step after

 Move before

 Move after

Extract previous...

View data source query

View query plan

 Properties

Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the graphic.

Answer Area

[Answer choice] of the transformation steps in the query will fold.

▼

All
None
Some

The Added custom step will be performed in **[answer choice]**.

▼

each lakehouse’s query engine
the Microsoft Power Query engine
the source lakehouse query engine

Answer(s): A

Answer Area

[Answer choice] of the transformation steps in the query will fold.

	▼
All	
None	
Some	

The Added custom step will be performed in [answer choice].

	▼
each lakehouse's query engine	
the Microsoft Power Query engine	
the source lakehouse query engine	

QUESTION

You have a Fabric tenant that contains a lakehouse named Lakehouse'. Lakehouse1 contains a table named Table1. You are creating a new data pipeline. You plan to copy external data to Table1. The schema of the external data changes regularly. You need the copy operation to meet the following requirements:

- Replace Table1 with the schema of the external data.
- Replace all the data in Table1 with the rows in the external data.
- You add a Copy data activity to the pipeline.

What should you do for the Copy data activity?

A.From the Source tab, add additional columns.

B.From the Destination tab, set Table action to Overwrite.

C.From the Settings tab, select Enable staging.

D.From the Source tab, select Enable partition discovery.

E.From the Source tab, select Recursively.

Answer(s): B

QUESTION

You have a Fabric tenant that contains a lakehouse. You plan to query sales data files by using the SQL endpoint. The files will be in an Amazon Simple Storage Service (Amazon S3) storage bucket. You need to recommend which file format to use and where to create a shortcut. Which two actions should you include in the recommendation?

A.Create a shortcut in the Files section.

B.Use the Parquet format

- C. Use the CSV format.
- D. Create a shortcut in the Tables section.
- E. Use the delta format.

Answer(s): D,E

QUESTION

You have a Fabric tenant that contains a lakehouse named Lakehouse1. Lakehouse1 contains a subfolder named Subfolder1 that contains CSV files. You need to convert the CSV files into the delta format that has V-Order optimization enabled. What should you do from Lakehouse explorer?

- A. Use the Load to Tables feature.
- B. Create a new shortcut in the Files section.
- C. Create a new shortcut in the Tables section.
- D. Use the Optimize feature.

Answer(s): A

QUESTION

You have a Fabric tenant that contains a lakehouse named Lakehouse1.

Lakehouse1 contains an unpartitioned table named Table1.

You plan to copy data to Table1 and partition the table based on a date column in the source data. You create a Copy activity to copy the data to Table1. You need to specify the partition column in the Destination settings of the Copy activity.

What should you do first?

A.From the Destination tab, set Mode to Append.

B.From the Destination tab, select the partition column.

C.From the Source tab, select Enable partition discovery.

D.From the Destination tabs, set Mode to Overwrite.

Answer(s): D

QUESTION

You have a Fabric tenant that contains a warehouse named Warehouse1.

Warehouse1 contains a fact table named FactSales that has one billion rows.

You run the following T-SQL statement.

CREATE TABLE test.FactSales AS CLONE OF Dbo.FactSales;
For each of the following statements, select Yes if the statement is true.
Otherwise, select No.

Answer Area

Statements	Yes	No
A replica of <code>dbo.Sales</code> is created in the test schema by copying the metadata only.	☐	☐
Additional schema changes to <code>dbo.FactSales</code> will also apply to <code>test.FactSales</code> .	☐	☐
Additional schema changes to <code>dbo.FactSales</code> will also apply to <code>test.FactSales</code> .	☐	☐

Answer(s): A

Answer Area

Statements	Yes	No
A replica of <code>dbo.Sales</code> is created in the test schema by copying the metadata only.	☒	☐
Additional schema changes to <code>dbo.FactSales</code> will also apply to <code>test.FactSales</code> .	☐	☒
Additional schema changes to <code>dbo.FactSales</code> will also apply to <code>test.FactSales</code> .	☒	☐

QUESTION

You have source data in a folder on a local computer. You need to create a solution that will use Fabric to populate a data store. The solution must meet the following requirements:

- Support the use of dataflows to load and append data to the data store.
- Ensure that Delta tables are V-Order optimized and compacted automatically.

Which type of data store should you use?

- A.a lakehouse
- B.an Azure SQL database
- C.a warehouse
- D.a KQL database

Answer(s): A

QUESTION

You have a Fabric tenant that contains a lakehouse. You are using a Fabric notebook to save a large DataFrame by using the following code.

```
df.write.partitionBy("year", "month",  
"day").mode("overwrite").parquet("Files/SalesOrder")
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Answer Area

Statements	Yes	No
The results will form a hierarchy of folders for each partition key.	☐	☐
The resulting file partitions can be read in parallel across multiple nodes.	☐	☐
The resulting file partitions will use file compression.	☐	☐

Answer(s): A

Answer Area

Statements	Yes	No
The results will form a hierarchy of folders for each partition key.	☒	☐
The resulting file partitions can be read in parallel across multiple nodes.	☒	☐
The resulting file partitions will use file compression.	☐	☒

QUESTION

You have a Fabric workspace named Workspace1 that contains a data flow named Dataflow1 contains a query that returns the data shown in the

following exhibit.

Table.TransformColumnTypes(#"Promoted headers", ({"VendorID", Int64.Type}, {"2015-01-01", Int64.Type}, {"2015-01-02", Int64.Type}, {"2015-01-03", Int64.Type}, {"2015-01-04", Int64.Type}, {"2015-01-05", Int64.Type}, {"2015-01-06", Int64.Type}, {"2015-01-07", Int64.Type}, {"2015-01-08", Int64.Type}))

	VendorID	2015-01-01	2015-01-02	2015-01-03	2015-01-04	2015-01-05	2015-01-06	2015-01-07	2015-01-08
1	1	16	15	0	9	12	9	9	9
2	2	20	15	7	17	20	6	6	6

Column profiling based on top 1,000 rows

Query settings >

Properties

Name
Generated-NYC-Taxi-Gre...

Entity type ⓘ
Custom

Applied steps

- Source
- Promoted ...
- Changed c...

Data destination +

No data destination

Step

Publish

You need to transform the data columns into attribute-value pairs, where columns become rows. You select the VendorID column.

Which transformation should you select from the context menu of the VendorID column?

A.Group by

B.Unpivot columns

C.Unpivot other columns

D.Split column

E.Remove other columns

Answer(s): C

QUESTION

You have a Fabric tenant that contains a data pipeline.You need to ensure that the pipeline runs every four hours on Mondays and Fridays. To what should you set Repeat for the schedule?

- A.Daily
- B.By the minute
- C.Weekly
- D.Hourly

Answer(s): C

QUESTION

You have a Fabric tenant that contains a warehouse. Several times a day, the performance of all warehouse queries degrades. You suspect that Fabric is throttling the compute used by the warehouse.

What should you use to identify whether throttling is occurring?

- A.the Capacity settings
- B.the Monitoring hub
- C.dynamic management views (DMVs)
- D.the Microsoft Fabric Capacity Metrics app

Answer(s): D

QUESTION

You have a Fabric workspace that uses the default Spark starter pool and runtime version 1.2. You plan to read a CSV file named Sales_raw.csv in a lakehouse, select columns, and save the data as a Delta table to the managed area of the lakehouse. Sales_raw.csv contains 12 columns. You have the following code.

```

from pyspark.sql.functions import year

(spark
 .red
 .format("csv")
 .option("header", 'true')
 .load("Files/sales_raw.csv")
 .select('SalesOrderNumber', 'OrderDate', 'CustomerName',
 'UnitPrice')
 .withColumn("Year", year("OrderDate"))
 .write
 .partitionBy('Year')
 .saveAsTable("sales")
 )

```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Answer Area

Statements	Yes	No
The Spark engine will read only the 'SalesOrderNumber', 'OrderDate', 'CustomerName', 'UnitPrice' columns from Sales_raw.csv.	☐	☐
Removing the partition will reduce the execution time of the query.	☐	☐
Adding inferSchema= 'true' to the options will increase the execution time of the query.	☐	☐

Answer(s): A

Answer Area

Statements	Yes	No
The Spark engine will read only the 'SalesOrderNumber', 'OrderDate', 'CustomerName', 'UnitPrice' columns from Sales_raw.csv.	☒	☐
Removing the partition will reduce the execution time of the query.	☐	☒
Adding <code>inferSchema= 'true'</code> to the options will increase the execution time of the query.	☒	☐

QUESTION

You have a Fabric tenant that contains a warehouse. A user discovers that a report that usually takes two minutes to render has been running for 45 minutes and has still not rendered. You need to identify what is preventing the report query from completing.

Which dynamic management view (DMV) should you use?

- A.sys.dm_exec_requests
- B.sys.dm_exec_sessions
- C.sys.dm_exec_connections
- D.sys.dm_pdw_exec_requests

Answer(s): A

QUESTION

You are creating a data flow in Fabric to ingest data from an Azure SQL database by using a T-SQL statement. You need to ensure that any foldable Power Query transformation steps are processed by the Microsoft SQL Server engine. How should you complete the code?

Values	Answer Area
EnableFolding	let
NativeQuery	Source = Sql.Databases("server.database.windows.net"
Optimize	),
Record	Database = Source{[Name = "db"]}[Data],
StopFolding	Query = [] . [] (
Table	Database,
Value	" SELECT * FROM customer WHERE country IN ('USA', 'UK')",
	null,
	[[] = true]
	)
	in
	Query

Answer(s): A

Answer Area

```
let
    Source = Sql.Databases(
        "server.database.windows.net"
    ),
    Database = Source{[Name = "db"]}[Data],
    Query = Value . NativeQuery (
        Database,
        " SELECT * FROM customer WHERE country IN ('USA', 'UK')",
        null,
        [ EnableFolding = true]
    )
in
    Query
```

QUESTION

You have a Fabric tenant that contains a lakehouse named Lakehouse1.

Readings from 100 IoT devices are appended to a Delta table in Lakehouse1.

Each set of readings is approximately 25 KB. Approximately 10 GB of data is received daily.

All the table and SparkSession settings are set to the default.

You discover that queries are slow to execute. In addition, the lakehouse storage contains data and log files that are no longer used.

You need to remove the files that are no longer used and combine small files into larger files with a target size of 1 GB per file.

What should you do?

Actions

Answer Area

Set the autoCompact
table setting.

Set the optimizeWrite
table setting.

Run the VACUUM
command on a schedule.

Set the autoCompact
SparkSession setting.

Run the OPTIMIZE
command on a schedule.

Set the parallelDelete
SparkSession setting.

Remove the files:

Combine the files:

Answer(s): A

Answer Area

Remove the files:

Run the VACUUM
command on a schedule.

Combine the files:

Run the OPTIMIZE
command on a schedule.

Question

You have a Fabric tenant that contains a workspace named Workspace1. Workspace1 contains a lakehouse. The lakehouse contains a table named Customers and a Fabric notebook.

You plan to use the notebook to profile the data.

In the notebook, you set up the following DataFrame that references the Customers table.

```
df = Spark.sql("Select * from Customers")
```

You need to profile the data in the DataFrame. The solution must minimize administrative effort.

What should you do?

A.Create a pandas DataFrame first, and then visualize the data in an embedded Microsoft Power BI report by running QuickVisualize method on the pandas DataFrame.

B.Display the DataFrame by running `display(df)`, and then clicking the **Inspect** button.

This answer is correct.

C.Display the DataDrame by running `display(df)`, and then switching to chart view.

D. Visualize the data in an embedded Microsoft Power BI report by running the QuickVisualize method on the DataFrame.

Answer(s): B

Explanation:

While you can embed a Power BI report in a notebook and analyze the data by using the QuickVisualize method, this does not generate the statistics required for profiling. The Chart option is a powerful tool to visualize data, but it does not provide statistics. The Inspect button provides summarized data distribution and statistics for each column.

[Notebook visualization - Microsoft Fabric | Microsoft Learn](#)

Question #

You have a Microsoft Power BI visual.

You use DAX query view to review the following code extracted from the visual.

DEFINE

```
    VAR __DS0Core =  
        SUMMARIZECOLUMNS('Company'[Manufacturer],  
"Sales", 'Metrics'[Sales], "ConstantSales",  
'Metrics'[ConstantSales])
```

```
VAR __DS0PrimaryWindowed =  
    TOPN(1001, __DS0Core, [Sales], 0,  
'Company'[Manufacturer], 1)
```

```
EVALUATE  
    ROW(  
        "AnalyticsLine",  
        MEDIANX(KEEPFILTERS(__DS0Core), [Sales])  
    )
```

```
EVALUATE  
    __DS0PrimaryWindowed
```

```
ORDER BY  
    [Sales] DESC, 'Company'[Manufacturer]
```

You need to identify which type of analytics line was added to the visual. What should you identify?

A. constant line

B. max line

C. median line

This answer is correct.

D.percentile line

Answer(s): C

Explanation:

In the code the section
containing `MEDIANX(KEEPFILTERS(__DS0Core),
[Sales])` indicates that there is a median line used in the visual.

[Use Performance Analyzer to examine report element performance in Power BI Desktop - Power BI | Microsoft Learn](#)

[DAX query view in Power BI Desktop - Power BI | Microsoft Learn](#)

Question #

You use Microsoft Power BI Desktop to connect to data stored in a CSV file. You need to use Power Query Editor to identify the percentage of valid records in a column before loading the data to a report. Which Power Query option should you use?

A.Column distribution

B.Column profile

C.Column quality

This answer is correct.

D. Filter by value

Answer(s): C

Explanation:

A percentage of valid records in the column is displayed when you enable Column quality. Column distribution provides an overview of the value frequency and distribution in a column. Column profile provides statistical data about values in a column.

Question #

You have a Fabric warehouse that contains the following tables:

- Sales (DateKey, ProductKey, SalesAmount)
- Product(ProductKey, ProductName)
- Date (DateKey, Day, Month, Year)

There is a 1-to-many relationship between the Date and Sales tables on DateKey. There is a 1-to-many relationship between the Product and Sales tables on ProductKey.

You begin to write the following SQL query to analyze the Sales data by ProductName and Year, but only for products that have a yearly SalesAmount of more than \$10000.

```
Select p.ProductName, d.Year, Sum(s.Sales Amount)
From Sales s
LEFT JOIN DimProduct p on s.ProductKey =
p.ProductKey
LEFT JOIN DimDate d on s.DateKey = d.DateKey
```

You need to complete the query to meet the requirements.

How should you complete the query?

A.GROUP BY p.ProductKey, d.DateKey HAVING
SUM(s.SalesAmount) > 10000]

B.GROUP BY p.ProductName, d.Year HAVING
SUM(s.SalesAmount) > 10000

This answer is correct.

C.WHERE s.SalesAmount > 10000 GROUP BY
p.ProductKey, d.DateKey

```
D.WHERE s.SalesAmount > 10000 GROUP BY  
p.ProductName, d.Year
```

Answer(s): B

Explanation:

The **GroupBy** columns must match the columns used in the **SELECT** statement. Using **WHERE** will eliminate individual sales records that has a daily **SalesAmount** that is larger than 10,000. The goal is to remove records for which the total **SalesAmount** for a year is larger than 10,000. This can be achieved by using **HAVING** since it works on the result of the **GroupBy**.

[HAVING \(Transact-SQL\) - SQL Server | Microsoft Learn](#)